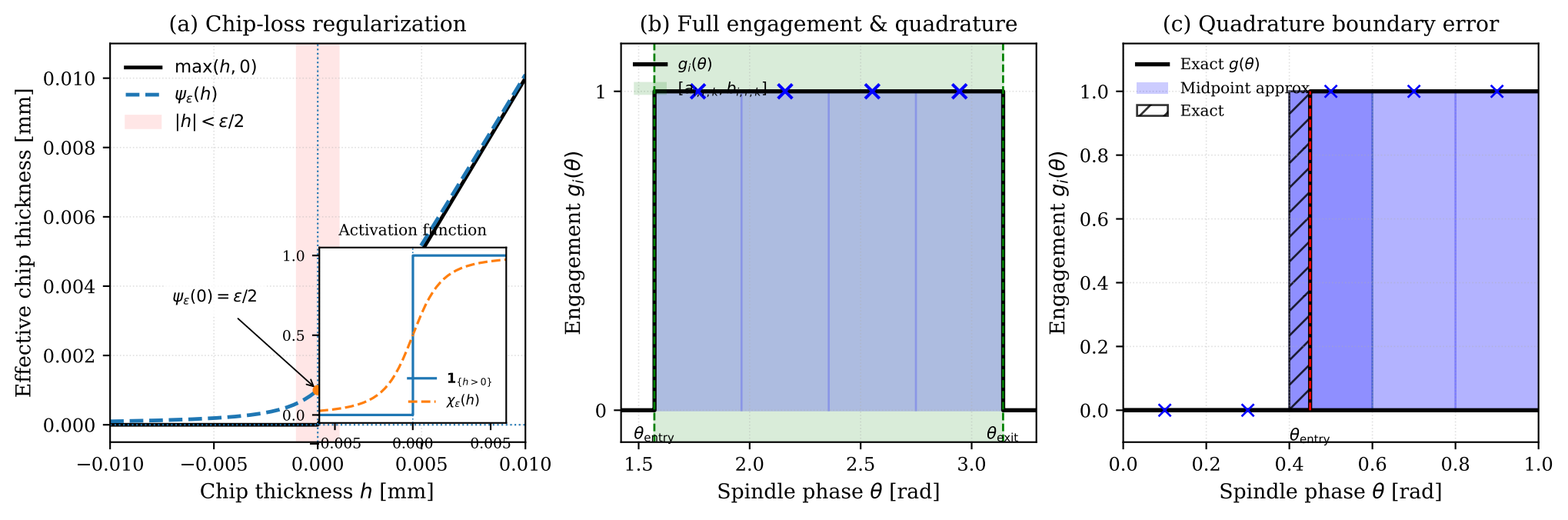
(a) Chip-loss regularization
max(h, 0)
ψε(h)
|h| < ε/2
Activation function
ψε(0) = ε/2
1{h>0}
χε(h)
Effective chip thickness [mm]
Chip thickness h [mm]
(b) Full engagement & quadrature
gi(θ)
θentry
θexit
Engagement gi(θ)
Spindle phase θ [rad]
(c) Quadrature boundary error
Exact g(θ)
Midpoint approx.
Exact
θentry
Engagement gi(θ)
Spindle phase θ [rad]